ELE00062I Acoustics and Studio Recording

Assessment Documentation

This document outlines the work you must complete for the assessment for the Acoustics and Studio Recording module. This consists of two pieces of work:

- Acoustic Analysis Exercise
- Studio Recording Exercise

Each of these is worth 50% of the module mark. The full brief for these exercises is presented below, along with the marking criteria and general guidance.

1. Acoustic Analysis Exercise

This section outlines the Acoustic Analysis Exercise, guidance on completing it, the deliverables you need to submit, and the marking criteria.

1.1. The assignment task

Your task is to prepare a report (and accompanying audio examples) covering an introduction to room acoustics and auralisation. To do this you will make use of an anechoic recording of a musical instrument, and an acoustic impulse response, to generate an auralisation. You should record the anechoic material yourself, and make use of an IR from the OpenAIR database (if you want to record an IR yourself then you will be provided with the tools required).

You will then conduct an analysis (using MATLAB) of the musical recording, the IR, and the auralisation, and will relate these analyses to the acoustic properties of the chosen musical instrument and the acoustic features of the impulse response measurement space. The results will be presented in a technical report, which should include:

- A brief introduction to room acoustics and IRs.
- A discussion of the physical characteristics of the IR recording space.
- A description and analysis of IR. This analysis should include the calculation of several parameters, including RT60, EDT, clarity, and definition.
- A critical evaluation of the anechoic recording, covering the acoustic properties of the chosen instrument, and the auralisation.
- Presentation of the IR, the recording, and the IR in graphical form using suitable time- and frequency-domain representations.
- A conclusion summarising the work presented and identifying at least three key points that demonstrate how the physical characteristics of the space result in the presence of particular acoustic features, and how these features affect the auralisation.

The report content should be aimed at an interested, non-expert, audience (a 1st-year MusTech student, for example). Ensure that everything you cover is clearly explained for the intended audience and supported with appropriate references.

1.2. Anechoic recording

To complete this assignment you will need to make an anechoic recording of a musical instrument of your choice. You do not have to be the performer that you record, and can conduct this recording in small groups if you wish (provided each student makes use of a different recording). To organise booking the anechoic chamber you must email elec-audiolab-tech@york.ac.uk. As an alternative you could make use of a totally dry synthesised sound source, though a full anechoic recording is preferred (and will receive more marks).

1.3. IR analysis (and measurement)

We will cover the theory and practice of IR measurement and analysis in lecture and lab sessions. These will provide you with the skills you need for this aspect of the assignment. You will conduct an analysis of your chosen IR according to the ISO 3382 standard[1]. You will also be expected to present time- and frequency-domain analyses of your recordings in the form of waveform plots, spectral plots, and spectrograms.

Your report should clearly present the results from these analyses and relate the findings to the physical properties of the space (i.e. its size, shape, dimensions, and surface materials). The IR you make use of should be in stereo. You will submit the IR, anechoic recording, and auralisation along with the MATLAB analysis code. It is perfectly fine for you to make use of an IR from OpenAIR[2], but you can record your own IR if you wish (it is your responsibility to organise access to the recording location).

If you do decide to record your own IR then you can book the equipment available through studiobooking.york.ac.uk. You can make such a recording in a small group and share the resultant IR(s), but your analysis work and your report must be your own individual work. If you decide to record your own IRs you must complete a risk assessment. There is a standard risk assessment for recording work on campus, which you must read and acknowledge before carrying out your work. If you wish to record in an off-campus, outdoor or otherwise potentially hazardous environment, this will need to be separately risk assessed before the work is carried out. Any enquiries can be sent to andrew.chadwick@york.ac.uk.

1.4. The report

You will also submit an individual report (maximum 6 pages, excluding appendices). Here is an outline of a typical report structure:

1. Introduction
2. Background information on the fundamentals of room acoustics and IRs.
3. Details of the IR recording space (size, shape, materials).
4. IR analysis:
    a. A description of the time and frequency domain features of the IR (supported by suitable plots)
    b. Calculation of room acoustic parameters according to ISO 3382 (including RT60, EDT, clarity, and definition)
5. A description of the anechoic recording work, and an analysis of the acoustic properties of the recorded instrument.
6. Critical evaluation of the auralisation
7. Conclusion
8. References
9. Appendices:

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[1] A copy of this standard will be available on the VLE page.
[2] https://openairlib.net/

a. Any tables of data or images which are too large to put in the main text.
b. Details of the accompanying audio files.
c. Descriptions of the submitted MATLAB code..

1.5. Your submission

You should upload the following to the VLE submission point in a single .zip file:

- Your report in PDF format.
- The IR (stereo), anechoic recording (mono), and auralisation (stereo).
- MATLAB analysis code. This should include a combination of code provided to you in the labs and your own code.

1.6. Marking

This task comprises **50% of the module mark** according to the following marking scheme:

Acoustic Analysis Exercise	
Criteria	**Weighting**
Report presentation and structure: - Structured well with a clear narrative and flow of information - Professional formatting - Easy to read, accurate, and suitable for the intended audience	**5%**
Technical content of the report: - Covers everything in the brief - A comprehensive description of the acoustic properties of the chosen IR. - Demonstration of a good understanding of underlying signal processing and acoustic theory. - Evidence of critical listening supported by objective analysis of the IR and auralisation. - Incisive evaluation of how the physical properties of the space relate to the acoustic features. - Suitable use of figures, tables, and diagrams.	**25%**
References: - A complete set of references including proper academic sources, correctly formatted.	**5%**
Accompanying files: - High quality anechoic recording, IR, and auralisation. - All files well labelled and curated, and submitted in the correct format. - Functional MATLAB scripts and functions allowing for recreation of the results presented in the report. Code is clear and well commented throughout.	**15%**
Total:	**50%**

2. Studio Recording Exercise

This section outlines the assignment task, guidance on completing it, the deliverables you need to submit, and the marking criteria.

For this part of the assignment, you are required to produce a studio-based multitrack recording. There are very few constraints on this task, but please bear in mind that your goal here is to demonstrate your skills in studio-based recording and production. Whatever you choose to record, make sure it allows you to showcase your abilities!

2.1. Multitrack Recording: The Track

There are no stipulations on the type of music you should record. You might create an original composition, or record a version of another artist's work. If you record an existing piece of music, you may choose to try to replicate the production of the original track as closely as possible, or record your own version in a new and creative way – the choice is yours.

The finished track should be **between 2:30 and 3:30 in length**. It should also contain several ***recorded*** elements. As a guideline, there should be at least 3 elements recorded with microphones, at least one of which should employ a stereo microphone technique covered on the course. You may, of course, use synths as part of your production, but ensure there are enough *recorded* elements to showcase that side of your production abilities.

You may not use any pre-recorded audio tracks/stems in your work. You cannot, for example, do a remix of a track based on parts released online, e.g. a vocal. The idea is that all recorded audio is your own work. You may use short audio/MIDI samples or loops from libraries, but **all external audio/MIDI used must be clearly referenced in the report**.

2.2. Multitrack Recording: The Recording Process

There are no rules on how you should put your multitrack together. You may choose to track a band in the studio all together, or assemble the elements piece by piece over several sessions. You do not have to record everything in the studio – you can record elements at home if you wish, or in other interesting-sounding spaces.

You may use as many effects, automation, or editing techniques you deem appropriate. You do not, however, *have* to use all of these things – a beautifully-recorded folk song, for example, will not lose marks because it doesn't make use of radical effects. Also bear in mind that the quality of your recordings is being assessed, and if your effects obscure this then this might be detrimental to your mark.

You are free to apply any mastering processing you wish to your finished stereo mix if you think it will enhance the result. Once you have a mix you are happy with, simply export it as a high-quality (44.1kHz, 16 bit, minimum 192kbps) mp3 for submission.

2.3. Multitrack Recording: The Report

Along with your multitrack recording, you should submit a report describing how you went about making it. It must be a **maximum of 4 pages including diagrams (single column, minimum of 11pt font, 2 cm margin).** Title/contents pages and references are not included in this count. The precise layout of the report is up to you, but you should address each of the following sections:

- **Choice of track**
  - Description of the piece
  - Aesthetic goals for the recording
  - Instrumentation/arrangement
- **The recording process**
  - Chosen approach (all-at-once, track-by-track etc.)
  - Order of recording the elements (if applicable)
  - Use of guide tracks or click tracks (or not)
- **Recording each element**
  - Microphone choice and placement for each instrument (you do not need to cover *everything*, but do give a good account of the most important/interesting elements)
  - Consideration of how each element is to fit into the mix, and any impact this had on recording choices
- **The mixing process**
  - Any editing performed on the takes
  - Effects applied to each element (EQ, compression, reverb, etc.)
  - Organising your project (any groups/bussing used during the mix)
  - Panning scheme for the track (diagrams help with this)
  - Any mastering effects applied to the finished mix (if applicable)
- **Evaluation**
  - Were you happy with the result?
  - Did it meet the creative aims?
  - Would you have done anything differently in terms of the recording process?

You do not have enough space to describe every single element of your mix in detail, so try to focus on the most unique aspects. Ensure you use diagrams and photos to illustrate your report – a good picture can be worth a thousand words in describing your setup.

2.4. Multitrack Recording: Deliverables

Your work is to be submitted using the **VLE**. The recordings should be encoded as high-quality (44.1kHz, minimum 192kbps) **mp3** files, while the accompanying reports should be saved as **PDF** files. Please include both of these files in a single ZIP file for upload through the VLE.

Submission must be anonymous, and is subject to the University rules on academic misconduct – all work must be your own, and any third-party material must be properly referenced.

2.5. Multitrack Recording: Marking Scheme

The Multitrack Recording accounts for **50% of your total module mark**, and is broken down into the following three categories:

- **Creative and Appropriate use of Recording Techniques (20% of module mark)**
    - Are the **Production Choices** appropriate to the track? This covers the overall presentation of the track, its creativity and sonic interest – the overall impression should be coherent, interesting, and appropriate to the style of music. The duration should be within the bounds set in the assignment.
    - Are the **Recorded Elements** good? There should be several *recorded* parts, at least one of which should employ a stereo microphone technique. Each recorded element should be clear and well presented, with any applied effects (whether subtle or extreme) being appropriate.
    - Does the track have good **Stereo Imaging**? The elements should be arranged (panned) to make the best possible use of the stereo field, ensuring that each one is clear and in its own space.
- **Technical Quality of the Track (20% of module mark)**
    - Is the **Production Quality** of the track good? The mix should sound clean and clear, with minimal noise and no artefacts (clicks or pops), and no clipping.
    - Is the **Dynamic Range** well managed? Overall, the track should have a good dynamic shape, and where compression has been applied it should be appropriate and not overdone.
    - Does the track have a well-managed **Frequency Range**? The mix should make full use of the audible spectrum, without any range lacking or being too forthright. Elements should sound full, but without masking each other, and the mix should not sound "muddy".
- **Written Report (10% of module mark)**
    - Is the **Writing and Presentation** of the report good? It should be clearly laid out, divided into sensible sections, and easy to read. Diagrams and figures should be labelled and referred to in the text, and external sources properly referenced.
    - **Documentation of the Process**. The report should clearly outline your aims/goals for the track, and the process you followed in completing it. The key recording and mixing techniques and processes should be clearly documented and justified.
    - Is there **Reflection and Evaluation** of the process and outcome? The report should critically evaluate your work and the resulting track. Were there things that could have been better? What did you learn along the way that might change the way you might do things in the future?

The following tables give representative statements for each of these factors in ranges of marks. Note that the mark awarded for each category will be an aggregate of the three columns.

Multitrack: Creative and Appropriate use of Recording Techniques (20% of module mark)

	Production Choices:	**Recorded Elements:**	**Stereo Imaging:**
17-20	Excellent production – a creative, coherent and clear piece. Recording/mixing choices present the music in the best light possible.	Excellent recordings, giving a clear presentation of each element. Stereo recording full and wide. Effects used well to enhance the recordings and flatter each element.	Excellent stereo imaging – mix is wide, balanced and full, with clearly discernible elements all having their own space.
13-16	Good production and a good result. Recording and mixing choices are appropriate to the style and piece, and maintain interest throughout.	Good quality recordings overall. Effects have been used well for the most part to enhance each element. Stereo recording present and effective.	Good stereo imaging. Elements are clearly placed, and the mix makes good use of the stereo field.
9-12	A fair production standard. Piece is listenable, but lacks variety. Timing/performance out in places. Too short/long.	Acceptable recordings, but mic/placement choices could have been better. Effects help somewhat, but may detract from elements in places. Too few recorded elements.	Reasonable use of the stereo field, some elements occasionally obscure each other, or mix is unbalanced in places.
5-8	Poor production – track lacks variety or sonic interest, and timing/performance is sloppy. Too short/long.	Recordings of poor quality, but elements are audible. Effects used in a limited capacity. Too few recorded elements.	Poor use of stereo imaging - mix is lopsided, cluttered or slightly unstable.
0-4	Incoherent or unfinished recording, with poor performance, production or mixing decisions. Far too short/long.	Elements are very poorly recorded and not clearly audible. Effects are either unused. No/few recorded elements.	Little or no stereo imaging – elements overlap and obstruct each other, or imaging is very unstable.

Multitrack: Technical Quality of the Track (20% of module mark)

	Overall Mix Quality:	**Dynamic Range:**	**Frequency Range:**
17-20	Crystal-clear mix and pristine presentation. Minimal noise and zero clicks/pops. Any editing is imperceptible.	Excellent dynamic range. Piece has a clear dynamic shape, and compression has been appropriately and skilfully applied.	Excellent, full mix with full bass, controlled mids and expansive high-end. Elements sound full, and fit together seamlessly.
13-16	Good-quality mix and presentation, with low noise and minimal audible artefacts.	A good dynamic range, with contrast between sections. Good use of compression, where applied.	A good mix, making full use of the frequency spectrum. Elements fit together without conflicting.
9-12	A fair quality mix. Some noise evident and the occasional click, pop or clip.	Some dynamic interest, but mix is lacking shape. Compression may be under- or over-used.	Reasonable use of the frequency range. Mix may be a little unbalanced or muddy, or elements may compete for space.
5-8	A poor mix, with audible noise and artefacts, and/or clipping.	Poor dynamics – mix is either heavily over-compressed or uncontrolled. Little or no meaningful contrast between sections.	Poor use of the frequency range. Mix/elements are very unbalanced, muddy, mask each other, or dominate one region of the spectrum.
0-4	Mix is extremely noisy and/or clipped. Audible and objectionable clicks/pops, edits or artefacts.	Very poor dynamic range – mix is entirely shapeless, and little or no effort has been made to manage dynamics.	Very poor frequency range – total lack of low or high frequencies, and muddy mid-range.

Multitrack: Written Report (10% of module mark)

	Writing and Presentation:	**Documentation of the Process:**	**Reflection and Evaluation:**
9-10	Excellent – well written and easy to read. Report is cleanly presented and clearly	Both goals and working process for the track are clearly justified and documented, giving a	Clear, full evaluation of both the process and outcome. Any problems or potential

	laid out. Sections are clear and sensibly ordered, diagrams properly labelled and referenced.	complete account of the creation of the track. Descriptions of recording/mixing techniques are clear, justified and informative.	improvements in either are clearly described.
7-8	A good report – well written, with clear sections. Diagrams are clear and referred to in the text.	Goals and working process are well described, and the recording/mixing techniques used are mostly well documented.	A good evaluation is given, and suggestions for improvement in process and outcome are made.
5-6	Report is fair – writing and presentation are OK, layout and formatting are acceptable. Referencing/labelling of diagrams may be missing in places.	A reasonable description of your goals and working process for the track. Description of the techniques used is unclear in places or omits important details.	Evaluation is fair, with some effort made to appraise process or outcome.
3-4	Poorly presented, and writing is unclear. Minimal attention to layout, formatting or referencing.	Unclear recording goals and poor description of the process. Limited description of the recording/mixing techniques used.	Minimal attempt made to evaluate process or outcome. No suggestions made for improvement.
0-2	Very poor – no effort to section or lay out the report, and rushed/uninformative writing. No diagrams or photos.	No useful description of the recording process or the techniques employed.	No meaningful evaluation given of either process or outcome.

3. General guidance

This section includes general information and guidance that applies to each of the two assignments for this module.

3.1. Support:

You should make sure that you start work on these assignments as early as possible and make use of the support available. You will have lab sessions dedicated to assignment work where you can receive guidance and formative feedback on your work.

3.2. Appendices:

You can add relevant extra material to your report in the form of appendices, which are not counted in the page limit. An example of this is the inclusion of images you want to include that are too large for the body of the report. Do not include code listings in your appendices: any code should be submitted as separate files.

3.3. Academic writing:

The library provides excellent resources presenting guidance on academic writing. You should make sure your writing is clear, concise, and correct.

3.4. References:

Your reports should clearly reference other work that has informed your own, or from which you have sources information and material. A list of these references should be included following the concluding section of your reports. The references should be in IEEE format.

3.5. LaTeX:

It is strongly advised that you use LaTeX to prepare your reports. An Overleaf template will be made available on the module's VLE page.

3.6. MATLAB Code

There are many repositories for MATLAB code available in books and/or online. The code you submit should reflect your own efforts, rather than that of any other person or third-party resource. If you do use some code, or coding ideas, from another source, you should clearly attribute where it has come from, in the code itself and in your references. Note that the more code you use from elsewhere, the fewer marks you will achieve, as it will not reflect your own effort and understanding. You may freely use any code supplied to you as part of the module teaching material.